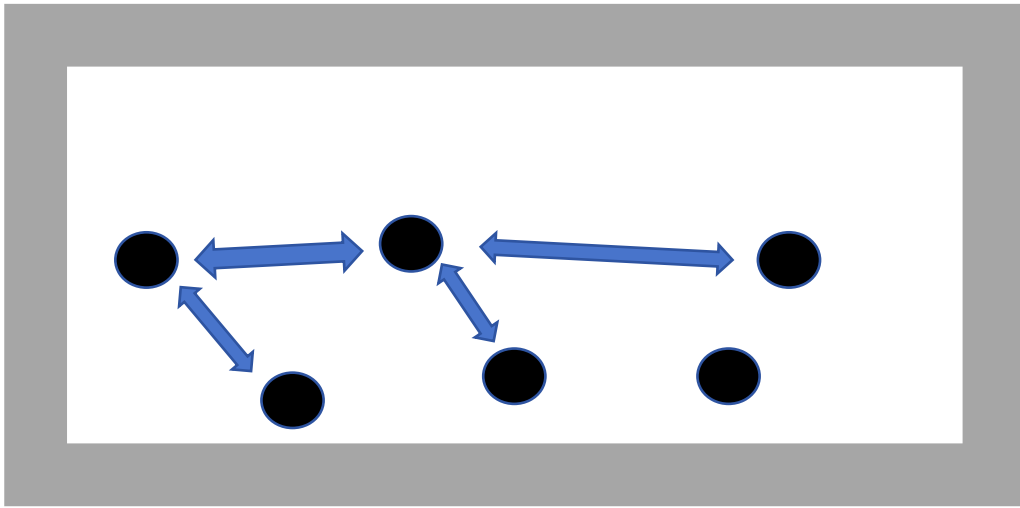


Current Algebra of HK Model

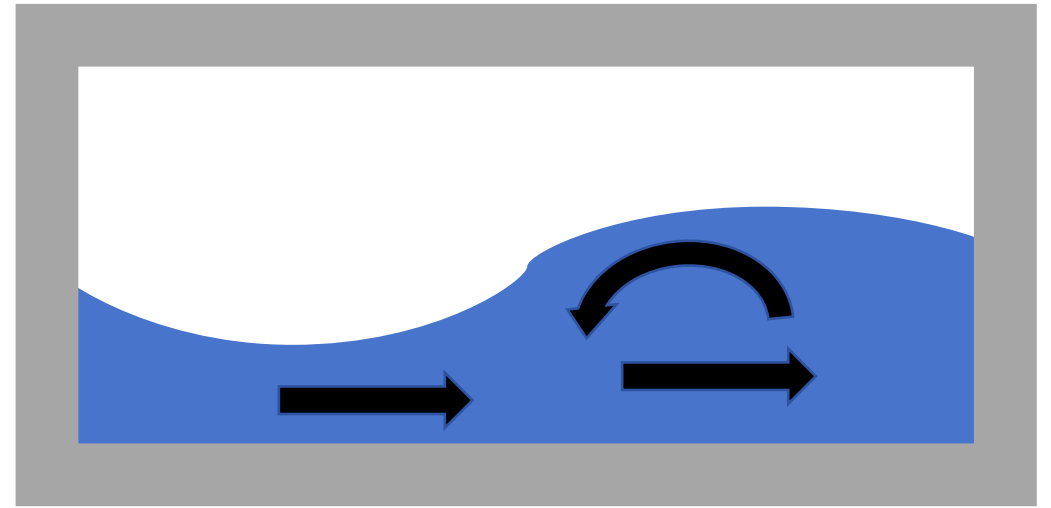
Yuting Bai

(1) Current Algebra Description of Free Fermion



Particle Description

$$c_{\mathbf{k}}, c_{\mathbf{k}}^{\dagger}$$



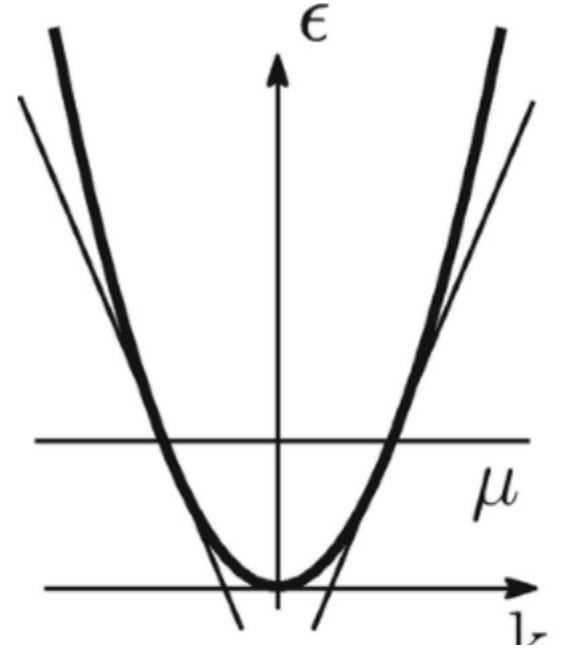
Fluid Description

$$\rho_k(q) = c_{k+q}^{\dagger} c_k$$

(1) Current Algebra of Free Fermion

$$H = \sum_{\mathbf{k}} \epsilon_{\mathbf{k}} n_{\mathbf{k}} \simeq \sum_{\mathbf{k}} v_F k (c_{\mathbf{kR}}^\dagger c_{\mathbf{kR}} - c_{\mathbf{kL}}^\dagger c_{\mathbf{kL}})$$

For free fermion, we check the operator dynamics in the Heisenberg picture.



Particle	Current
$c_{\mathbf{kR}}(t) = e^{-iv_F k t} c_{\mathbf{kR}}$	$\rho_{qR}(t) = \sum_{\mathbf{k}} e^{iv_F(k+q)t} c_{\mathbf{k+qR}}^\dagger c_{\mathbf{kR}} e^{-iv_F k t}$
$c_{\mathbf{kR}}^\dagger(t) = e^{iv_F k t} c_{\mathbf{kR}}^\dagger$	$= e^{iv_F q t} \rho_{qR}$

Both of them acquire a U(1) phase under evolution!

$$\rho_q \equiv \frac{1}{L} \int dx e^{iqx} \rho(x) = \sum_{\mathbf{k}} c_{\mathbf{k+q}}^\dagger c_{\mathbf{k}}$$

(1)Current Algebra of Free Fermion

- Getting the Equation of Motion correct is not enough.
- Does current operators ρ_q obey boson's commutation relation?

$$\begin{aligned} & [\rho_{\mathbf{q},\mathbf{a}}, \rho_{\mathbf{q}',\mathbf{b}}] \\ &= \frac{\delta_{\mathbf{ab}}}{N} \sum'_{\mathbf{k}_1, \mathbf{k}_2} [c_{\mathbf{q}+\mathbf{k}_1,\mathbf{a}}^\dagger c_{\mathbf{k}_1,\mathbf{b}}, c_{\mathbf{q}'+\mathbf{k}_2,\mathbf{a}}^\dagger c_{\mathbf{k}_2,\mathbf{b}}] \\ &= \frac{\delta_{\mathbf{ab}}}{N} \sum'_{\mathbf{k}_1, \mathbf{k}_2} (\delta_{\mathbf{q}'+\mathbf{k}_2,\mathbf{k}_1} c_{\mathbf{q}+\mathbf{k}_1,\mathbf{a}}^\dagger c_{\mathbf{k}_2,\mathbf{b}} - \delta_{\mathbf{q}+\mathbf{k}_1,\mathbf{k}_2} c_{\mathbf{q}'+\mathbf{k}_2,\mathbf{a}}^\dagger c_{\mathbf{k}_1,\mathbf{b}}) \\ &= \frac{\delta_{\mathbf{ab}}}{N} \sum'_{\mathbf{k}_1} (c_{\mathbf{q}+\mathbf{k}_1,\mathbf{a}}^\dagger c_{-\mathbf{q}'+\mathbf{k}_1,\mathbf{b}} - c_{\mathbf{q}+\mathbf{q}'+\mathbf{k}_1,\mathbf{a}}^\dagger c_{\mathbf{k}_1,\mathbf{b}}) \stackrel{?}{=} 0 \end{aligned} \quad \longrightarrow \quad \text{Do we fail?}$$

(1)Current Algebra of Free Fermion

- The canonical commutation relation does not give you the right answer!

$$\frac{\delta_{ab}}{N} \sum'_{\mathbf{k}_1} (c_{\mathbf{q}+\mathbf{k}_1,\mathbf{a}}^\dagger c_{-\mathbf{q}'+\mathbf{k}_1,\mathbf{b}} - c_{\mathbf{q}+\mathbf{q}'+\mathbf{k}_1,\mathbf{a}}^\dagger c_{\mathbf{k}_1,\mathbf{b}})$$

The summation is performed on the interval $[-\Lambda, \Lambda]$



$$\frac{\delta_{ab}}{N} \left(\sum_{\mathbf{k}_1=-\Lambda-\mathbf{q}'}^{-\Lambda} c_{\mathbf{q}+\mathbf{q}'+\mathbf{k}_1,\mathbf{a}}^\dagger c_{\mathbf{k}_1,\mathbf{b}} - \sum_{\mathbf{k}_1=\Lambda-\mathbf{q}'}^{\Lambda} c_{\mathbf{q}+\mathbf{q}'+\mathbf{k}_1,\mathbf{a}}^\dagger c_{\mathbf{k}_1,\mathbf{b}} \right)$$



$$[\rho_{\mathbf{q},\mathbf{a}}, \rho_{\mathbf{q}',\mathbf{b}}] = \delta_{ab} \operatorname{sgn}(\mathbf{a}) \frac{L\mathbf{q}}{2\pi} \delta_{\mathbf{q}+\mathbf{q}',0}$$

The commutator of current operator is a c-number. It is boson!

(2) Bjorken-Johnson-Low Prescription

- The previous method works well for free fermion. Here we introduce a method that calculates the anomalous commutator directly from correlation function.

$$T(\omega) = \int dt e^{i\omega t} \langle a | T A(t) B(0) | b \rangle$$

$$\lim_{\omega \rightarrow \infty} \omega T(\omega) = i \langle a | [A(0), B(0)]_- | b \rangle .$$

The matrix element of anomalous commutator depends on the high-frequency limit. This implies the robustness of anomalous commutator to interaction!

(2) Bjorken-Johnson-Low Prescription

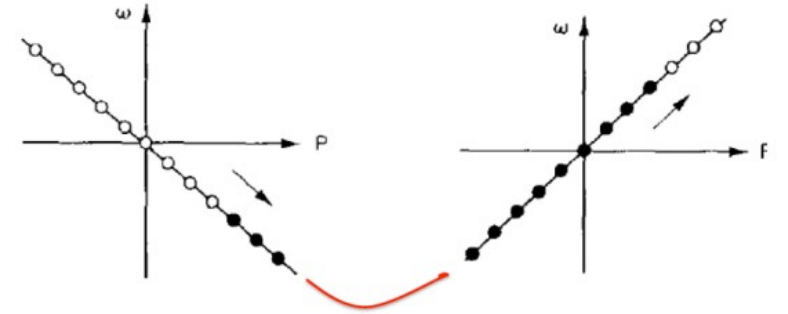
- One can show that BJL prescription reproduce the U(1) Kac Moody algebra for chiral fermion.

$$\begin{aligned}
 T(\omega) &= \delta_{ab} \delta_{\mathbf{q}+\mathbf{q}',0} \int dt e^{i\omega t} \sum_{\mathbf{k}_1, \mathbf{k}_2} \delta_{\mathbf{k}_1+\mathbf{q}, \mathbf{k}_2} e^{-i \text{sgn}(\mathbf{a}) v_F q t} f_a(\mathbf{k}_1, \mathbf{k}_2) \\
 &= i \delta_{ab} \delta_{\mathbf{q}+\mathbf{q}',0} \sum_{\mathbf{k}_1} \left(\frac{n_{\mathbf{k}_1+\mathbf{q},a} (1 - n_{\mathbf{k}_1,a})}{\omega - \text{sgn}(\mathbf{a}) v_F q + i0^+} - \frac{n_{\mathbf{k}_1,a} (1 - n_{\mathbf{k}_1+\mathbf{q},a})}{\omega - \text{sgn}(\mathbf{a}) v_F q - i0^+} \right)
 \end{aligned} \tag{20}$$



$$\lim_{\omega \rightarrow \infty} \omega T(\omega)$$

$$\langle 0 | [n_{\mathbf{q},a}, n_{\mathbf{q}',b}] | 0 \rangle = i \delta_{ab} \delta_{\mathbf{q}+\mathbf{q}',0} \sum_{\mathbf{k}_1} [n_{\mathbf{k}_1+\mathbf{q},a} (1 - n_{\mathbf{k}_1,a}) - n_{\mathbf{k}_1,a} (1 - n_{\mathbf{k}_1+\mathbf{q},a})]$$



(3)HK Model

- HK Model: A NFL theory that breaks Z_2 symmetry.

$$H = \sum_{\mathbf{k}\sigma} \epsilon_{\mathbf{k}} n_{\mathbf{k}\sigma} + U \sum_{\mathbf{k}} n_{\mathbf{k}\uparrow} n_{\mathbf{k}\downarrow}$$

- The HK model has Green function zero, which leads to violation of Luttinger Theorem.

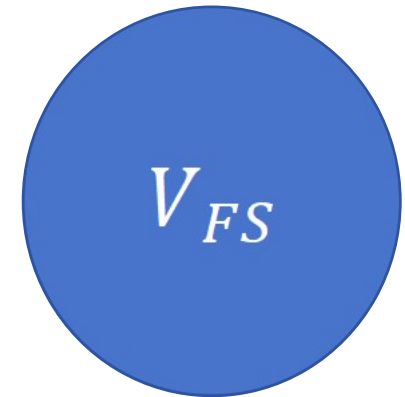
Could one find a local solvable theory that has the similar correlation function with HK in the low-energy, long wavelength limit?

$O(4)$



~~Z_2~~ $\times SO(4)$

$N_{tot} \neq$



(4)Current Algebra for HK

- In HK model, the eigenparticles are not bare fermions, but holon and doublon!

$$c_{k\sigma}(t) = e^{-i\epsilon_k t} c_{k\sigma} (1 - n_{k\bar{\sigma}}) + e^{-i(\epsilon_k + U)t} c_{k\sigma} n_{k\bar{\sigma}}$$

- Instead of bare current, one should work with parton current

$$\rho_{q,a\sigma} = \sum_{\alpha,\beta=\xi,\eta} \rho_{q,a\sigma}^{\alpha\beta}$$

$$\rho_{q,a\sigma}^{\alpha\beta} = \sum_k (c_{k+q,a\sigma}^{\alpha})^{\dagger} c_{k,a\sigma}^{\beta}$$

$$c_{k\sigma}^{\xi} = c_{k\sigma} P_{k\bar{\sigma}}^{\xi} = c_{k\sigma} (1 - n_{k\bar{\sigma}})$$

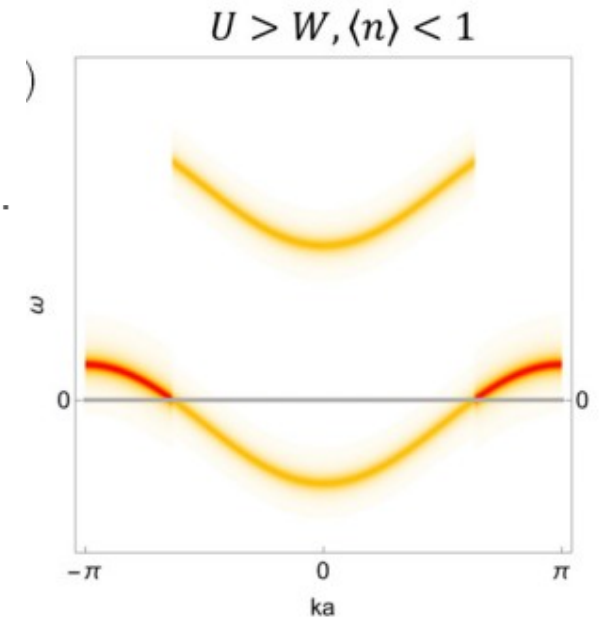
$$c_{k\sigma}^{\eta} = c_{k\sigma} P_{k\bar{\sigma}}^{\eta} = c_{k\sigma} n_{k\bar{\sigma}}$$

$\rho^{\xi\xi}, \rho^{\eta\eta}$: Projection of current to Upper/Lower Hubbard Band

$\rho^{\xi\eta}, \rho^{\eta\xi}$: Mixture of Upper/Lower Hubbard Band

(4) Current Algebra of HK

- We may use B JL prescription to calculate anomalous commutators between these parton currents
 - We consider ground state with filling surface at Low Hubbard Band
- (1) Since no double occupancy, there is no support for J^{UU}
- (2) We assume the perturbation is small enough that one can **approximate the commutator with the ground state average value.**
- (3) We average over all degenerate ground state in HK when evaluate commutator using B JL



(4)Current Algebra of HK

- We may start with projected density $\rho^{\xi\xi}$.

$$[\rho_{qa}^{\xi\xi}, \rho_{q'a'}^{\xi\xi}] = -\frac{\delta_{aa'}}{2} \text{sgn}(a) \delta_{q+q',0} \frac{qL}{2\pi}$$

- In general, the projected current $\rho^{\xi\xi}$ is not conserved subjected to an external field.

The Projected Charge

$$N_L = N - \sum_{\mathbf{k}} n_{\mathbf{k}\uparrow} n_{\mathbf{k}\downarrow}$$



Add Onsite
Potential



$$H = H_{HK} + \sum_{\mathbf{k}\sigma} V_{-\mathbf{k}\sigma} \rho_{\mathbf{k}\sigma}$$

$$[N_L, H] = - \sum_{\mathbf{k}\sigma} V_{-\mathbf{k}\sigma} (\rho_{\mathbf{k}\sigma}^{\eta\xi} - \rho_{\mathbf{k}\sigma}^{\xi\eta})$$

Must include J^{LU} , J^{UL} to
satisfy conservation law!!

(4)Current Algebra of HK

$$\left[\rho_{qa}^{\xi\xi}, \rho_{q'a'}^{\xi\xi} \right] = -\frac{\delta_{aa'}}{2} \text{sgn}(a) \delta_{q+q',0} \frac{qL}{2\pi}$$

$$\left[\rho_{qa}^{\xi\xi}, \rho_{q'a'}^{\xi\eta} \right] = \delta_{aa'} \rho_{q+q'a}^{\xi\eta}$$

$$\left[\rho_{qa}^{\xi\xi}, \rho_{q'a'}^{\eta\xi} \right] = -\delta_{aa'} \rho_{q+q'a}^{\eta\xi}$$

$$\left[\rho_{qa}^{\eta\xi}, \rho_{q'a'}^{\xi\eta} \right] = -\frac{\delta_{aa'}}{2} \left(\rho_{q+q'a}^{\xi\xi} + \frac{1}{2} \text{sgn}(a) \delta_{q+q',0} \frac{qL}{2\pi} \right),$$

The 1st/4th line is derived from B JL prescription.
The 2nd/3rd line is fixed by conservation law.

To get the commutator $[\rho^{\eta\xi}, \rho^{\xi\eta}]$, there are two ways
(1)using the Jacobi identity.
(2)using B JL.
The latter yields an extra nonlocal equal-time commutator, which is suppressed by $\frac{q}{k_F}$.



(4) Current Algebra of HK

The parton current in the charge sector is affine
 $\mathfrak{su}(2)$ Lie algebra!

$$\left[\varrho_{qa\sigma}^i, \varrho_{q'a'\sigma'}^j \right] = \delta_{aa'} \delta_{\sigma\sigma'} \left(i \epsilon_{ijk} \varrho_{q+q'a}^k - \frac{\text{sgn}(a)}{2} \delta_{q+q',0} \frac{qL}{2\pi} \right)$$

$$\varrho^z = \rho^{\xi\xi}$$

$$\varrho^x = \frac{1}{2} (\rho^{\xi\eta} + \rho^{\eta\xi})$$

$$\varrho^y = \frac{1}{2i} (\rho^{\xi\eta} - \rho^{\eta\xi})$$

(5) Hamiltonian

- Under the Heisenberg picture, the current operator only gain a phase under time evolution.

$$\begin{aligned}\rho_{qa\sigma}^{\xi\xi}(t) &= e^{i\epsilon_{qa}t} \rho_{qa\sigma}^{\xi\xi} \\ \rho_{qa\sigma}^{\xi\eta}(t) &= e^{i(\epsilon_{qa}-U)t} \rho_{qa\sigma}^{\xi\eta} \\ \rho_{qa\sigma}^{\eta\xi}(t) &= e^{i(\epsilon_{qa}+U)t} \rho_{qa\sigma}^{\eta\xi}\end{aligned}$$



$$[H(\rho_{qa\sigma}^{\alpha\beta}), \rho_{qa\sigma}^{\alpha\beta}] = E_{qa\sigma}^{\alpha\beta} \rho_{qa\sigma}^{\alpha\beta}$$

(5) Hamiltonian

- We can use a LOCAL hamiltonian to generate the correct energy.

$$H = v_F \frac{2\pi}{L} \sum_{i=x,y,z} \sum_{qa\sigma} \varrho_{qa\sigma}^i \varrho_{-qa\sigma}^i - U \sum_{a\sigma} \varrho_{q=0a\sigma}^z.$$



J_z commutes with J^2 if there is no anomaly. Schwinger term gives you kinetic energy $v_F q$.



Zeeman field that generates the energy difference U

(5) Hamiltonian

- We can use a LOCAL hamiltonian to generate the correct energy.

$$H = v_F \frac{2\pi}{L} \sum_{i=x,y,z} \sum_{qa\sigma} \varrho_{qa\sigma}^i \varrho_{-qa\sigma}^i - U \sum_{a\sigma} \varrho_{q=0a\sigma}^z.$$



$$H = 2\pi v_F \sum_{a\sigma} \int dx \vec{\varrho}_{a\sigma}^2(x) - \sum_{a\sigma} \int dx U \varrho_{a\sigma}^z(x).$$

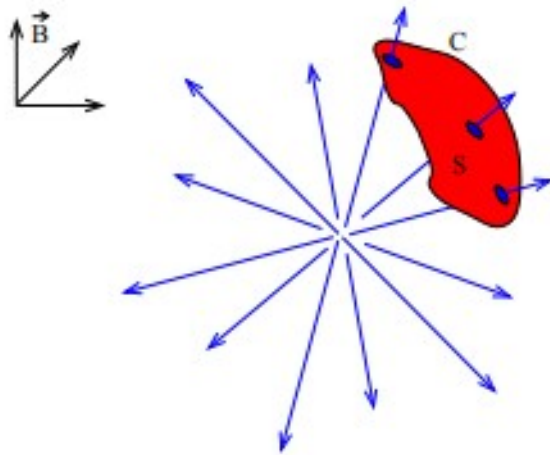
(6) Topology and Geometry Aspect

- We could modify the second term.

$$H[\mathbf{U}(t)] = \sum_{a\sigma} \int dx [2\pi v_F \vec{\varrho}_{a\sigma}^2(x) - \mathbf{U}(x, t) \cdot \vec{\varrho}_{a\sigma}(x)].$$



Berry Phase



$$S_{\text{eff}}[\vec{U}] = c \int dt \int_0^1 ds \vec{U} \cdot (\partial_t \vec{U} \times \partial_s \vec{U}) + \dots$$

The effective action
must be **multi-valued**
function of $\mathbf{U}$!

Relation with anomaly
in LW Functional?

(6) Correlation function

- We may compare correlation function from our Hamiltonian and the Kubo formalism

Kubo : Imaginary time-ordered correlation function

$$\langle \rho \rho \rangle (q, \omega) = \left(\rho - \frac{|q|}{2\pi} \right) \frac{U}{\omega^2 - U^2} + \frac{v_F q^2}{\pi \omega^2} + o_{q \rightarrow 0}(q^2)$$

Current Algebra: Real time retarded correlation function

$$\langle \rho \rho \rangle_B^R (q, \omega) = \frac{\rho U}{\omega^2 - U^2} + \frac{v_F q^2}{\pi \omega^2} + O(q^2)$$

The difference is controlled by $\frac{q}{k_F}$ and $\frac{v_F q}{U}$.

In the limit $k_F, U \rightarrow \infty$, the difference goes to zero.

(7) New Interpretation for HK model

- We show that a local model could restore the equation of motion and two-body correlation of band HK model at IR limit.
- The local, well-defined excitation in HK model are particle-hole pair instead of single particle operators!
- Maybe we could interpret the nonlocality of HK model as a result of rewriting local degree of freedom in nonlocal way.

$$(U_x + iU_y)\rho_{qa\sigma}^{\xi\eta} \Leftrightarrow \sum_k (U_x + iU_y)(c^\xi)_{k+qa\sigma}^\dagger c_{ka\sigma}^\eta,$$



Local SU(2) Object



Nonlocal particle-hole cloud

(11) To Do List

- (1) Ambiguity of BJJ Prescription
- Replacing commutator with ground state expectation value might miss something

$$\begin{aligned}
 [\rho_{qa}^{\xi\xi}, \rho_{q'a'}^{\xi\xi}] &= -\frac{\delta_{aa'}}{2} \text{sgn}(a) \delta_{q+q',0} \frac{qL}{2\pi} \\
 [\rho_{qa}^{\xi\xi}, \rho_{q'a'}^{\xi\eta}] &= \delta_{aa'} \rho_{q+q'a}^{\xi\eta} \\
 [\rho_{qa}^{\xi\xi}, \rho_{q'a'}^{\eta\xi}] &= -\delta_{aa'} \rho_{q+q'a}^{\eta\xi} \\
 [\rho_{qa}^{\eta\xi}, \rho_{q'a'}^{\xi\eta}] &= -\frac{\delta_{aa'}}{2} \left(\rho_{q+q'a}^{\xi\xi} + \frac{1}{2} \text{sgn}(a) \delta_{q+q',0} \frac{qL}{2\pi} \right),
 \end{aligned}$$


 These terms have zero expectation value on GS, while necessary to close the algebra!

- (2) Response: How to couple the Hamiltonian to gauge field
- (3) Generalization to other model: MMHK, 1D Hubbard, higher dimension?

Thank You!